

A UNIFORM BOGOMOLOV TYPE RESULT FOR TAUTOLOGICAL CYCLES IN JACOBIANS OF CURVES IN ONE-PARAMETER FAMILIES.

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1. MAIN THEOREM

Let K be any number field, X any smooth projective geometrically connected curve of genus $g \geq 2$, $J = \text{Pic}^0(X)$ its Jacobian, $\alpha \in \text{Div}^1(X)$ any divisor of degree 1 and $m \in (\mathbb{Z} \setminus \{0\})^r$ for some $1 \leq r \leq g-1$ any vector of non-zero integers. We denote a generalized Abel–Jacobi map by

$$f_{m,\alpha}: X^r \rightarrow J, \quad (x_1, \dots, x_r) \mapsto \sum_{j=1}^r m_j(x_j - \alpha)$$

and we write $Z_{m,\alpha}$ for the image of $f_{m,\alpha}$. Further, we write $(n)_m = \prod_{k=0}^{m-1} (n-k)$.

Theorem 1. *Let $\pi: \mathcal{X} \rightarrow S$ be a smooth morphism, where S is a quasi-projective curve defined over K and all fibers $\pi^{-1}(t) = \mathcal{X}_t$ are smooth projective geometrically connected curves of genus $g \geq 2$. Let $1 \leq r \leq g-1$ and $m \in (\mathbb{Z} \setminus \{0\})^r$. Then there exist constants c_1 and $c_2 > 0$ such that for all $X \in \{\mathcal{X}_t \mid t \in S(\overline{K})\}$ and all $\alpha \in \text{Div}^1(X)$ there exists a divisor $Z \subseteq Z_{m,\alpha}$ whose degree is bounded by*

$$3 \max \left(4^{2g+3r} r! (g)_r \prod_{j=1}^r m_j^2, 4^{3g+r} (4e)^{r+6} g^{\frac{c_2 - c_1 + 65g \log g}{4c_2}} \right)^2 r! (g)_r \prod_{j=1}^r m_j^2$$

and which contains all geometric points $P \in Z_{m,\alpha}(\overline{K})$ satisfying

$$h_{NT}(P) \leq \frac{(r+1)c_2}{g(4e)^{r+6}}.$$

Remark 2. (i) *Everything is explicit except for c_1 and c_2 , which are defined by minima of certain C^∞ -functions on $\mathcal{M}_g(\mathbb{C})$. It is a purely complex analytic problem to make them explicit.*

(ii) *We obtain the following uniform Bogomolov type result as a special case: For any $\alpha \in \text{Div}^1(X)$ let $j_\alpha: X \rightarrow J$, $P \mapsto P - \alpha$ be the Abel–Jacobi embedding. Then it holds*

$$\# \left\{ P \in X(\overline{K}) \mid h_{NT}(P) \leq \frac{2c_2}{g(4e)^7} \right\} \leq 3 \cdot 4^{2g+3} g^3 \max \left(1, \frac{c_2(g) - c_1(g) + 65g \log g}{4^3 c_2(g)} (4e)^7 \right)$$

for all $X \in \{\mathcal{X}_t \mid t \in S(\overline{K})\}$ and $\alpha \in \text{Div}^1(X)$.

(iii) *In contrast to the statement in my arXiv-paper we can only handle one-parameter families. I overlooked the existence of holomorphic maps $\Delta^* = \{0 < |t| < 1\} \rightarrow \mathcal{M}_g(\mathbb{C})$ which do not continue to continuous maps $\Delta = \Delta^* \cup \{0\} \rightarrow \overline{\mathcal{M}}_g(\mathbb{C})$.*

(iv) *The case of higher-dimensional bases S (and hence uniformity for the entire moduli space $\mathcal{M}_g$) is a joint work in progress with Robin de Jong and Farbod Shokrieh.*

2. GLOBAL COMPONENTS OF THE PROOF

2.1. The work of David–Philippon. Let $Y \subseteq J$ be a subvariety of dimension d which is no translate of an abelian subvariety. Then there exists a divisor $Z \subseteq Y$ with $\deg(Z) \leq \frac{\delta^2}{3 \cdot 16^{2g}} \deg(Y)$ with

$$\delta = 3 \cdot 16^2 \cdot \max \left(1, \frac{12(4e)^{d+2}(d+1)(4^{g+1}h_\Theta(J) + 3g \log 2)}{\hat{h}(Y)} \right) \deg(Y)$$

containing all $P \in Y(\overline{K})$ with $\hat{h}(P) \leq \frac{\hat{h}(Y)}{(4e)^{d+2} \deg(Y)}$. Here, $h_\Theta(J)$ denotes the (Arakelov)-height of $\Theta(J)$ for the theta embedding $\Theta: J \rightarrow \mathbb{P}^{16g-1}$ and $\hat{h}$ is the normalized height associated to $\mathcal{L} = \Theta^* \mathcal{O}(1)$. In particular $\hat{h}(P) = 16h_{NT}(P)$. Our goal is to compare the heights $\hat{h}(Z_{m,\alpha})$ and $h_\Theta(J)$ with the Faltings height $h_{\text{Fal}}(J)$ such that it can be cancelled out in the formula for δ .

2.2. The work of Pazuiki. The theta height $h_\Theta(J)$ can be compared to the stable Faltings height $h_{\text{Fal}}(J)$ by

$$h_\Theta(J) - \frac{1}{2}h_{\text{Fal}}(J) - \frac{1}{4[K:\mathbb{Q}]} \sum_{\sigma: K \rightarrow \mathbb{C}} \log \det(\text{Im } \tau_\sigma) \leq \frac{g}{2} \log 2\pi + g \log 4 + \log c_g,$$

where c_g denotes a bound for the θ -function, which can be set to $c_g = \frac{g+2}{2}$ for $g \leq 3$ and $c_g = \frac{g+2}{2} \left(\frac{g+2}{\pi\sqrt{3}} \right)^{g/2}$ for $g \geq 4$ due to Autissier. The matrix $\tau_\sigma \in \mathcal{F}_g$ is chosen in the Siegel fundamental space such that $J \otimes_\sigma \mathbb{C} \cong \mathbb{C}^g / (\mathbb{Z}^g + \tau_\sigma \mathbb{Z}^g)$. From this, one can deduce $h_\Theta(J) \leq h_{\text{Fal}}(X) + 5.2g \log g$.

2.3. Computing heights of tautological cycles. By definition it holds

$$\hat{h}(Y) = \frac{(\hat{\mathcal{L}}|_Y)^{d+1}}{[K:\mathbb{Q}]} \quad \text{and} \quad \deg Y = (\hat{\mathcal{L}}|_Y)^d,$$

where $\hat{\mathcal{L}}$ is $\mathcal{L} = \Theta^* \mathcal{O}(1)$ equipped with its admissible adelic metric and the powers mean (arithmetic) intersection products. By the projection formula we have $(\hat{\mathcal{L}}|_{Z_{m,\alpha}})^{r+1} = (f_{m,\alpha}^* \hat{\mathcal{L}})^{r+1}$ and one can show that

$$f_{m,\alpha}^* \hat{\mathcal{L}} = -8 \sum_{j,k=1}^r m_j m_k \hat{\Delta}_{jk}^\alpha$$

where $\hat{\Delta}^\alpha = \hat{\Delta} - p_1^* \hat{\alpha} - p_2^* \hat{\alpha} + \pi_2^* \hat{\alpha}^2$ denotes the canonical admissible adelic metrized line bundle on X^2 for the projections p_1, p_2 and structure morphism $\pi_2: X^2 \rightarrow \text{Spec } K$ and $\hat{\Delta}_{jk}^\alpha = p_{jk}^* \hat{\Delta}^\alpha$ for the projections $p_{jk}: X^r \rightarrow X^2$. We define the following correspondence

$$\begin{aligned} & n\text{-tuple } (\hat{\Delta}_{j_1 k_1}^\alpha, \dots, \hat{\Delta}_{j_n k_n}^\alpha) \text{ up to order} \\ \leftrightarrow & \text{ Graph } \Gamma = \Gamma(\hat{\Delta}_{j_1 k_1}^\alpha, \dots, \hat{\Delta}_{j_n k_n}^\alpha) : V = \{0, \dots, r\}, E = \{(j_1, k_1), \dots, (j_n, k_n)\}. \end{aligned}$$

We write $\langle \Gamma \rangle$ for the (arithmetic) intersection number $\hat{\Delta}_{j_1 k_1}^\alpha \cdot \dots \cdot \hat{\Delta}_{j_n k_n}^\alpha$. It satisfies the following properties

- (a) $\langle \Gamma \rangle = \prod_{\Gamma_j \subseteq \Gamma} \langle \Gamma_j \rangle$, where the product runs over all connected components of Γ .
- (b) If there exists $v \in V$ with $\deg v \leq 1$, then $\langle \Gamma \rangle = 0$.
- (c) If $v \in V$ has degree $\deg v = 2$ and $(v, v) \notin E$, we write $\Gamma/\{v\}$ for the graph obtained by contracting one edge incident to v . Then $\langle \Gamma \rangle = \langle \Gamma/\{v\} \rangle$.

(d) After these reduction steps, the remaining non-trivial cases are

$$\left\langle \bullet \circlearrowright \right\rangle = -2g,$$

$$\left\langle \circlearrowleft \bullet \circlearrowright \right\rangle = \frac{g}{g-1} \hat{\omega}^2 + 4(g-1)h_{NT}(x_\alpha),$$

$$\left\langle \circlearrowleft \bullet \text{---} \bullet \circlearrowright \right\rangle = -4(g-1)^2 h_{NT}(x_\alpha),$$

$$\left\langle \bullet \text{---} \bullet \right\rangle = \frac{2g+1}{2g-2} \hat{\omega}^2 - \varphi(X) + 6(g-1)h_{NT}(x_\alpha),$$

where $x_\alpha = \alpha - \frac{\omega}{2g-2}$, $\hat{\omega}$ denotes the canonical admissible adelic metrized canonical bundle on X and $\varphi(X) = \hat{\omega}^2 - \hat{\Delta}^3$, which is always positive due to Zhang and Cinkir.

By counting all relevant graphs, one can show

$$\deg Z_{m,\alpha} = 16^r r! (g)_r \prod_{j=1}^r m_j^2, \quad \frac{\hat{h}(Z_{m,\alpha})}{(r+1) \deg Z_{m,\alpha}} = \frac{8(g-r)}{[K:\mathbb{Q}]} (a \hat{\omega}^2 + b \varphi(X) + c h_{NT}(x_\alpha)),$$

where

$$a = \frac{\sum_{j=1}^r m_j^2}{4(g-1)^2} - \frac{(2g+1) \sum_{j \leq k} m_j m_k}{6g(g-1)^2(g-2)}, \quad b = \frac{\sum_{j \leq k} m_j m_k}{3g(g-1)(g-2)}, \quad c = \frac{(\sum_{j=1}^r m_j)^2}{g}.$$

2.4. Lower bound for $\hat{\omega}^2$. If one takes $m = (1, 1)$, $\alpha = \frac{\omega}{2g-2}$ and $\overline{\mathcal{M}} = \hat{\Delta}_{11}^\alpha + \hat{\Delta}_{22}^\alpha + 2g\hat{\Delta}_{12}^\alpha$, then the arithmetic Hodge index theorem for adelic line bundles by Yuan-Zhang gives

$$0 \geq f_{m,\alpha}^* \hat{\mathcal{L}} \cdot \overline{\mathcal{M}} = 64g^2 \left(\varphi(X) - \frac{2g+1}{g-1} \hat{\omega}^2 \right).$$

Equivalently, we obtain the lower bound $\hat{\omega}^2 \geq \frac{g-1}{2g+1} \varphi(X)$.

2.5. Zhang's invariants φ . To understand better the invariant $\varphi(X)$, we compute

$$\hat{\Delta}^3 = (\hat{\Delta}|_\Delta)^2 - \sum_{v \in M(K)} \int_{X(\overline{K}_v)^2} \log \|1\|_{\Delta,v} c_1(\hat{\Delta})^2 = \hat{\omega}^2 - \sum_{v \in M(K)} \int_{X(\overline{K}_v)^2} g_v c_1(\hat{\Delta})^2,$$

where $M(K)$ denotes the set of places of K and g_v is the Arakelov-Green function on the Riemann surface X_v for infinite places v or else on the reduction graph associated to $X_v = X \otimes_K K_v$. Thus, $\varphi(X)$ can be decomposed into a sum of invariants $\varphi(X_v)$ of Riemann surfaces and invariants $\varphi(\Gamma(X_v))$ of reduction graphs

$$\varphi(X) = \sum_{v \in |\text{Spec}(\mathcal{O}_K)|} \varphi(\Gamma(X_v)) \log N_v + \sum_{\sigma: K \rightarrow \mathbb{C}} \varphi(X_\sigma).$$

3. METRIZED GRAPHS

A metrized graph is a compact and connected metric space Γ , which is locally around any point $p \in \Gamma$ isometric to the star-shaped set

$$S(n_p, r_p) = \{z \in \mathbb{C} \mid z = te^{2\pi i k/n_p} \text{ for some } 0 \leq t \leq r_p \text{ and } k \in \mathbb{Z}\},$$

for some integer $n_p \geq 1$ and some real number $r_p > 0$, or to a point, in which case we set $n_p = 0$. Denote $\delta(\Gamma)$ for the total length of Γ .

The tangent space $T_p(\Gamma)$ of Γ at $p \in \Gamma$ is given by

$$T_p(\Gamma) = \{\gamma: [0, \epsilon] \rightarrow \Gamma \text{ path in } \Gamma \text{ with } \gamma(0) = p\} / \sim,$$

where $\gamma_1 \sim \gamma_2$ if there exists $\epsilon > 0$ such that $\gamma_1 \equiv \gamma_2$ on $[0, \epsilon]$. For $v \in T_p(\Gamma)$, represented by a path γ , we define

$$d_v f(p) = \lim_{t \rightarrow 0^+} \frac{f(\gamma(t)) - f(p)}{t}$$

if it exists and the set of functions

$$\text{Zh}(\Gamma) = \{f: \Gamma \rightarrow \mathbb{R} \mid f \text{ continuous, piecewise } C^2 \text{ and } d_v f(p) \text{ exists for all } p \in \Gamma, v \in T_p(\Gamma)\}.$$

Zhang defined the Laplace operator for $f \in \text{Zh}(\Gamma)$ by

$$\Delta(f) = f'' dx + \sum_{p \in \Gamma} \left(\sum_{v \in T_p(\Gamma)} d_v f(p) \right) \delta_p.$$

Here, δ_p is the Dirac measure at p .

For any measure μ of volume 1 on Γ there exists a unique continuous and symmetric function $g_\mu: \Gamma \times \Gamma \rightarrow \mathbb{R}$, called the Green function associated to μ , such that $g_\mu(x, \cdot) \in \text{Zh}(\Gamma)$ is characterized by

$$\Delta_y g_\mu(x, y) = \delta_y - \mu \quad \text{and} \quad \int_{\Gamma} g_\mu(x, y) \mu(y) = 0$$

for all $x \in \Gamma$. A polarization on Γ is an effective divisor $K = \sum_{p \in \Gamma} (n_p + m_p - 2)p$ with $m_p \geq 0$. For any function $f: \Gamma \rightarrow \mathbb{R}$ and any divisor $D = \sum_{p \in \Gamma} m_p p$ on Γ we write $f(D) = \sum_{p \in \Gamma} m_p f(p)$.

There exists a unique measure μ of volume 1 on Γ such that $g_\mu(x, K) + g_\mu(x, x)$ is independent of $x \in \Gamma$. We call $g = g_\mu$ the canonical Green function of (Γ, K) . Zhang's ϵ -invariant is given by

$$\epsilon(\Gamma, K) = \int_{\Gamma} g(x, x) ((2g - 2)\mu + \delta_K)$$

and his φ -invariant by

$$\varphi(\Gamma, K) = -\frac{1}{4}\delta(\Gamma) + \frac{1}{4} \int_{\Gamma} g(x, x) ((10g + 2)\mu - \delta_K).$$

It follows from the work by Cinkir, that

$$0 \leq \delta(\Gamma) + \epsilon(\Gamma, K) \leq \frac{23g^2 + 11g + 2}{(g-1)^2} \varphi(\Gamma, K),$$

for $g = \frac{1}{2} \deg K + 1$ the genus of the polarized metrized graph (Γ, K) .

4. RIEMANN SURFACES

Let M be any compact and connected Riemann surface of genus $g \geq 2$ and τ a period matrix, such that $J_M = \mathbb{C}^g / (\mathbb{Z}^g + \tau\mathbb{Z}^g)$ is its Jacobian. The pullback of the canonical 1-1 form

$$\nu = \frac{i}{2} \sum_{j,k=1}^g ((\operatorname{Im} \tau)^{-1})_{jk} dz_j \wedge d\bar{z}_k$$

on J_M by the Abel–Jacobi map $I: M \rightarrow J_M$ gives the canonical 1-1 form $\mu = \frac{1}{g} I^* \nu$ on M . The Arakelov–Green function $G: M^2 \rightarrow \mathbb{R}_{\geq 0}$ is characterized by

$$\partial_x \bar{\partial}_x \log G(x, y) = \pi i (\mu(x) - \delta_y(x)), \quad \int_M \log G(x, y) \mu(x) = 0.$$

The Zhang–Kawazumi invariant is given by

$$\varphi(M) = \int_{M^2} (\log G) c_1(\Delta)^2 > 0.$$

The Riemann theta function is given by

$$\theta: \mathbb{H}_g \times \mathbb{C}^g \rightarrow \mathbb{C}, \quad \theta(\tau, z) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i {}^t n \tau n + 2\pi i {}^t n z).$$

We associate to it a real-valued version $\|\theta\|: \mathbb{H}_g \times \mathbb{C}^g \rightarrow \mathbb{R}_{\geq 0}$ by

$$\|\theta\|(z) = \|\theta\|(\tau, z) = \det(\operatorname{Im} \tau)^{1/4} \exp(-\pi {}^t (\operatorname{Im} z) (\operatorname{Im} \tau)^{-1} (\operatorname{Im} z)) \cdot |\theta|(z),$$

which only depends on the class of z in J_M and which is invariant under the action of $\operatorname{Sp}_{2g}(\mathbb{Z})$. The I -invariant is given by

$$\begin{aligned} I(J_M) &= \frac{1}{2} \log \int_{J_M} \|\theta\|^2(z) \frac{\nu^g(z)}{g!} - \frac{1}{2} \int_{J_M} \log \|\theta\|^2(z) \frac{\nu^g(z)}{g!} \\ &= -\frac{g \log 2}{4} - \int_{J_M} \log \|\theta\|(z) \frac{\nu^g(z)}{g!} > 0. \end{aligned}$$

Finally, we may define Faltings δ -invariant by the formula

$$\delta(M) = 24I(J_M) + 2\varphi(M) - 2g \log 2\pi^4.$$

5. THE ARITHMETIC NOETHER FORMULA

Going back to the notation in the first two sections, we may assume that X has semi-stable reduction over $B = \operatorname{Spec}(\mathcal{O}_K)$ after a finite extension of K and we write $\rho: \mathcal{X} \rightarrow B$ for its minimal regular model. The stable Faltings height is defined by

$$h_{\text{Fal}}(X) = \frac{1}{[K: \mathbb{Q}]} \widehat{\deg} \det \rho_* \omega_{\mathcal{X}/B},$$

where $\omega_{\mathcal{X}/B}$ is the relative dualizing sheaf, $\det \rho_* \omega_{\mathcal{X}/B}$ is equipped with its canonical metric induced by the inner product $\langle \omega, \omega' \rangle = \frac{i}{2} \int \omega \wedge \bar{\omega}'$ on $H^0(X_{\mathbb{C}}, \Omega_{X_{\mathbb{C}}}^1)$ and $\widehat{\deg}$ denotes the Arakelov degree.

The vertex set V of the reduction graph $\Gamma_v(X)$ of X at $v \in |B|$ consists of the irreducible components of the special fibre $\mathcal{X}_v$ and its edge set consists of the singular points in $\mathcal{X}_v$, such that any edge e connects the two not necessarily different vertices associated to the irreducible components meeting in the singular point corresponding to e . It is a metrized graph by equipping every edge with the length 1 and it is polarized by the divisor

$$K = \sum_{p \in V} (n_p + g_p - 2)p,$$

where g_p denotes the genus of the normalization of the irreducible component of $\mathcal{X}_v$ corresponding to $p \in V$.

If we set

$$\delta(X) = \sum_{v \in |B|} (\delta(\Gamma_v(X)) + \epsilon(\Gamma_v(X))) \log N(v) + \sum_{v: K \rightarrow \mathbb{C}} (\delta(X_v) - 4g \log 2\pi)$$

where $N(v) = \#\mathcal{O}_K/v\mathcal{O}_K$, then the arithmetic Noether formula states

$$12[K : \mathbb{Q}]h_{\text{Fal}}(X) = \hat{\omega}^2 + \delta(X).$$

Using the inequalities for the invariants of reduction graphs, it remains only to find an inequality of the form

$$(1) \quad \varphi(M) \geq \frac{(g-1)^2}{23g^2+11g+2} \max(\delta(M) + c_1, c_2)$$

with $c_2 > 0$ for all $M \in \{\pi^{-1}(t) \mid t \in S(\mathbb{C})\}$ to prove the main theorem.

6. POLARIZED REAL TORI

An Euclidean lattice is a pair $(\Lambda, [\cdot, \cdot])$, where Λ is a finitely generated free $\mathbb{Z}$ -module and

$$[\cdot, \cdot]: \Lambda_{\mathbb{R}} \times \Lambda_{\mathbb{R}} \rightarrow \mathbb{R}$$

is an inner product on $\Lambda_{\mathbb{R}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$. A polarized real torus is the compact Riemannian manifold $\Sigma = \Lambda_{\mathbb{R}}/\Lambda$ associated to $(\Lambda, [\cdot, \cdot])$.

The tropical Riemann theta function associated to Σ is defined by

$$\Psi: \Lambda_{\mathbb{R}} \rightarrow \mathbb{R}, \quad \Psi(\nu) = \min_{\lambda \in \Lambda} \left\{ \frac{1}{2}[\lambda, \lambda] + [\nu, \lambda] \right\}.$$

This can be modified to an Λ -invariant function descending to Σ

$$\|\Psi\|: \Sigma \rightarrow \mathbb{R}, \quad \|\Psi\|(\nu) = \Psi(\nu) + \frac{1}{2}[\nu, \nu] = \min_{\lambda \in \Lambda} \frac{1}{2}[\nu + \lambda, \nu + \lambda].$$

The tropical moment of Σ is given by

$$I(\Sigma) = 2 \int_{\Sigma} \|\Psi\| \mu_H,$$

where μ_H denotes the Haar measure on Σ with volume 1.

Example 3. (a) To any positive-definite symmetric matrix $B \in \mathbb{R}^{g \times g}$ we associate an Euclidean lattice $B\mathbb{Z}^g = (B\mathbb{Z}^g, [\cdot, \cdot]_B)$, where the inner product is induced by $[Bn, Bn']_B = {}^t n B n'$ for any $n, n' \in \mathbb{Z}^g$. Let $\Sigma_B = \mathbb{R}^g/B\mathbb{Z}^g$ be the associated polarized real torus. For $x \in \mathbb{R}^g$ we have

$$\|\Psi_B\|(Bx) = \min_{Bn \in B\mathbb{Z}^g} \frac{1}{2}[Bx + Bn, Bx + Bn]_B = \frac{1}{2} \min_{n \in \mathbb{Z}^g} {}^t(x + n)B(x + n).$$

Further, the tropical moment of Σ_B is

$$I(\Sigma_B) = 2 \int_{x \in [0,1]^g} \|\Psi\|(Bx) \lambda(x),$$

where λ denotes the Lebesgue measure on $\mathbb{R}^g$.

(b) If V and $E = \{e_1, \dots, e_r\}$ denote the vertex and edge set of a model of a metrized graph Γ equipped with any orientation, the metric on Γ defines an inner product on

$$H_1(\Gamma, \mathbb{R}) = H_1(\Gamma, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{R} = \ker \left(\bigoplus_{i=1}^r \mathbb{Z} e_i \xrightarrow{\delta} \bigoplus_{v \in V} \mathbb{Z} v \right) \otimes_{\mathbb{Z}} \mathbb{R}$$

by $[e_i, e_j] = \delta_{ij} \ell(e_i)$. The tropical Jacobian is defined as the polarized real torus $J_{\Gamma} = H_1(\Gamma, \mathbb{R})/H_1(\Gamma, \mathbb{Z})$. It has been shown by de Jong and Shokrieh, that

$$\delta(\Gamma) + \epsilon(\Gamma, K) = 12I(J_{\Gamma}) + 2\varphi(\Gamma, K).$$

7. DEGENERATIONS

Let $f: \mathcal{X} \rightarrow \Delta$ be any smooth family of semistable complex projective curves of genus $g \geq 2$ over the complex unit disc Δ , where any fibre over $\Delta^* = \Delta \setminus \{0\}$ is smooth. We denote Γ_f for the polarized metrized reduction graph at $t = 0$. It has been proven by de Jong, that

$$\delta(\mathcal{X}_t) \sim -(\delta(\Gamma_f) + \epsilon(\Gamma_f)) \log |t| - 6g_0(\Gamma_f) \log(-\log |t|),$$

where $g_0(\Gamma)$ denotes the first Betti number of the graph Γ and $\sim$ means, that the difference of both side extends to a continuous function on Δ . Our goal is to prove the following similar result for the I -invariant.

Theorem 4. *With the notation above it holds*

$$2I(J_{\mathcal{X}_t}) \sim -I(J_{\Gamma_f}) \log |t| - \frac{g_0(\Gamma_f)}{2} \log(-\log |t|).$$

As a consequence, we obtain the asymptotic behaviour

$$\varphi(\mathcal{X}_t) \sim -\varphi(\Gamma_f) \log |t|.$$

Proof of Inequality (1). Let $\bar{S}$ be the closure of the natural image of $S(\mathbb{C})$ in $\overline{\mathcal{M}}_g(\mathbb{C})$. As $\bar{S}$ is compact and $\varphi(\mathcal{X}_t) > 0$ for all $t \in S$, it is enough to show, that $\varphi(\mathcal{X}_t)$ is arbitrary large locally around any point in $\bar{S} \setminus S$ and that

$$\frac{23g^2+11g+2}{(g-1)^2} \varphi(\mathcal{X}_t) - \delta(\mathcal{X}_t)$$

is bounded from below locally around any point in $\bar{S} \setminus S$. It is also enough to study this locally on the components of the normalization of $\bar{S}$, such that we can assume that $\bar{S}$ is smooth.

To show the first part, we choose for any $p \in \bar{S} \setminus S$ an open neighbourhood $\Delta \rightarrow \bar{S}$ which maps 0 to p and consider the family $f: \mathcal{X} \rightarrow \Delta$ induced by the pullback of $\pi: \mathcal{X} \rightarrow \bar{S}$. Then by its asymptotic behaviour, $\varphi(\mathcal{X}_t)$ is arbitrary large locally around p , as $\varphi(\Gamma_f) > 0$. The second part follows in the same way, as we also have

$$\frac{23g^2+11g+2}{(g-1)^2} \varphi(\Gamma_f) - (\delta(\Gamma_f) + \epsilon(\Gamma_f)) \geq 0.$$

□

8. PROOF OF THEOREM 4

The family $f: \mathcal{X} \rightarrow \Delta$ induces a multi-valued period map $T_f: \Delta^* \rightarrow \mathbb{H}_g$ with

$$T_f(t) = \begin{pmatrix} S_1(t) & S_3(t) \\ {}^t S_3(t) & \frac{\log t}{2\pi i} B + S_2(t) \end{pmatrix},$$

where $S_1: \Delta \rightarrow \mathbb{H}_{g_1}$, $S_3: \Delta \rightarrow \mathbb{C}^{g_1 \times g_2}$ and $S_2: \Delta \rightarrow \mathbb{C}^{g_2 \times g_2}$ are single-valued holomorphic maps for some integers $g_1 + g_2 = g$ and $B \in \mathbb{Z}^{g_2 \times g_2}$ is a symmetric and positive-definite matrix. We define the polarized real torus associated to f as $\Sigma_f = \Sigma_B$.

For any section $z: \Delta^* \rightarrow J_{\mathcal{X}}$ we define

$$\text{trop}(z) \equiv -\lim_{t \rightarrow 0} \frac{2\pi \text{Im}(\tilde{z}_2(t))}{\log |t|} \bmod B\mathbb{Z}^{g_2} \in \Sigma_f,$$

where $\tilde{z} = (\tilde{z}_1, \tilde{z}_2): \Delta^* \rightarrow \mathbb{C}^{g_1} \times \mathbb{C}^{g_2}$ is a multi-valued holomorphic map lifting z . If z has a lift $\tilde{z}: \Delta^* \rightarrow \mathbb{C}^g$ of the form $\tilde{z}(t) = a + T_f(t)b$ for two real-valued vectors $a, b \in \mathbb{R}^g$ with $b = (b_1, b_2) \in \mathbb{R}^{g_1} \times \mathbb{R}^{g_2}$, we get

$$\|\Psi_f\|(\text{trop}(z)) = \|\Psi_f\|(Bb_2) = \frac{1}{2} \min_{n \in \mathbb{Z}^{g_2}} {}^t(b_2 + n)B(b_2 + n).$$

Proposition 5. *With the notation as above, there exists a non-zero real analytic function $\alpha: \mathbb{R}^{2g} \rightarrow \mathbb{R}_{\geq 0}$, such that*

$$\lim_{t \rightarrow 0} \left(\|\theta\|(T_f(t), z(t)) \cdot |t|^{-\|\Psi_f\|(\text{trop}(z))} (-\log |t|)^{-g_2/4} \right)^2 = \alpha(a, b).$$

Proof. As $a + T_f(t)b$ represents $z(t)$, by an elementary calculation we have

$$\det(\text{Im } T_f(t))^{-1/4} \|\theta\|(T_f(t), z(t)) = \left| \sum_{n \in \mathbb{Z}^g} \mathbf{e} \left(\frac{1}{2} {}^t(n+b)T_f(t)(n+b) + {}^t n a \right) \right|.$$

If we put in the definition of T_f as a block matrix, we can factor out the t -power

$$t^{\frac{1}{2} \min_{n \in \mathbb{Z}^{g_2}} {}^t(n+b_2)B(n+b_2)} = t^{\|\Psi_f\|(\text{trop}(z))}$$

and we can show, that the remaining factor is bounded and not identical to 0.

Further, one can directly compute

$$\lim_{t \rightarrow 0} \frac{\det(\text{Im } T_f(t))}{(-\log |t|)^{g_2}} = \left(\frac{1}{2\pi}\right)^{g_2} \det(\text{Im } S_1(0)) \det(B) > 0.$$

□

As $I(J_{\mathcal{X}_t})$ and $I(\Sigma_f)$ are defined by the integrals of $\|\theta\|$ and $\|\Psi_f\|$, we get

$$\lim_{t \rightarrow 0} \left(I(J_{\mathcal{X}_t}) + \frac{g \log 2}{4} + \frac{1}{2} I(\Sigma_f) \log |t| + \frac{g_2}{4} \log(-\log |t|) \right) = -\frac{1}{2} \int_{[0,1]^{2g}} \log \alpha(a, b) \lambda(a, b).$$

Thus, Theorem 4 follows by the following two lemmas.

Lemma 6. *For any non-zero analytic function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ the function $\log |f|$ is locally integrable.*

By Hironaka's resolution theorem it is enough to check this if f is locally around $q \in \mathbb{R}^n$ given by $f = \epsilon \prod_{j=1}^n x_j^{k_j}$ for some $k_j \in \mathbb{Z}_{\geq 0}$ and an invertible analytic function ϵ .

Lemma 7. *It holds $\Sigma_f \cong J_{\Gamma_f}$.*

This lemma follows by constructing a basis of $H_1(\mathcal{X}_0, \mathbb{Z})$ by the union of a basis $B_1, \dots, B_r$ of $H_1(\Gamma_f, \mathbb{Z})$ and a symplectic basis of $H_1(\tilde{\mathcal{X}}_0, \mathbb{Z})$, where $\tilde{\mathcal{X}}_0$ denotes the normalization of the special fibre $\mathcal{X}_0$, which deforms to a symplectic basis of $H_1(\mathcal{X}_t, \mathbb{Z})$ for a general fibre $\mathcal{X}_t$. Computing the period matrix associated to this basis, one sees, that the multi-valued part B has to be given by the entries

$$B_{ij} = \sum_{e \in E(\Gamma_f)} \ell(e) b_{j,e} b_{i,e},$$

where the numbers $b_{i,e} \in \{-1, 0, 1\}$ are the coefficients of the edges in the basis elements $B_1, \dots, B_r$:

$$B_i = \sum_{e \in E(\Gamma_f)} b_{i,e} e.$$

Now one finds out, that the map

$$\alpha: H_1(\Gamma_\pi, \mathbb{Z}) \rightarrow B\mathbb{Z}^{g_2}, \quad \sum_{i=1}^{g_2} c_i B_i \mapsto B^t(c_1, \dots, c_{g_2})$$

is an isomorphism of Euclidean lattices, as

$$\begin{aligned} [B_i, B_j]_{\Gamma_f} &= \left[\sum_{e \in E(\Gamma_f)} b_{i,e} e, \sum_{e \in E(\Gamma_f)} b_{j,e} e \right] = \sum_{e \in E(\Gamma_f)} b_{i,e} b_{j,e} \ell(e) = B_{ij} \\ &= {}^t \alpha(B_i) B^{-1} \alpha(B_j) = [\alpha(B_i), \alpha(B_j)]_B. \end{aligned}$$